

Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Date	18 th June 2026
Project Name	ShopEZ
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Functional Requirements:

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	Registration through Registration Form
		Registration through Google (Gmail)
FR-2	User Authentication	User Login using Email & Password
		JWT Authentication
		Forgot Password (Optional)
FR-3	Product Management	View Products
		Search Products
		Filter Products by Category
		View Product Details
FR-4	Shopping Cart	Add Product to Cart
		Update Cart Quantity
		Remove Product from Cart
FR-5	Order Management	Place Order
		Order Confirmation
		View Order History
		Track Order Status
FR-6	Payment Processing	Select Payment Method
		Secure Online Payment
		Cash on Delivery (Optional)
FR-7	User Profile	View Profile
		Update Profile

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
		Change Password
FR-8	Admin Dashboard	Admin Login
		Manage Products
		Manage Categories
		Manage Users
		Manage Orders
		View Sales Analytics

Non-functional Requirements:

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The application should provide a simple, intuitive, and responsive user interface that enables users to browse, search, and purchase products with ease.
NFR-2	Security	User passwords should be encrypted using bcrypt, and secure authentication and authorization should be implemented using JWT. All sensitive data must be protected from unauthorized access.
NFR-3	Reliability	The system should consistently perform all shopping operations, including user authentication, cart management, and order processing, without failures.
NFR-4	Performance	Product pages, search results, and API responses should load quickly, ensuring a smooth shopping experience even with multiple users.
NFR-5	Availability	The ShopEZ platform should be available 24×7 with minimal downtime to ensure uninterrupted online shopping services.
NFR-6	Scalability	The MERN Stack architecture should support future enhancements such as payment gateway integration, recommendation systems, mobile applications, and increased user traffic.
NFR-7	Maintainability	The project should follow the MVC architecture and modular coding practices, making it easier to maintain, debug, and extend.
NFR-8	Compatibility	The application should function correctly across modern web browsers such as Chrome, Edge, Firefox, and Safari and support different screen sizes.